

15 GRAMS

THE AMOUNT OF HEMOGLOBIN PER LITER OF HEALTHY HUMAN BLOOD

one **giant continent**, which he named Gondwanaland, before the seas rose and they were separated. He was right about the giant continent, but we now know that they separated through continental drift (see 1915) and not because the seas rose.

At Langenltheim in Germany, the **first almost complete fossil of *Archaeopteryx*** was found. This winged, feathered creature with reptilian teeth, which lived 150 million years ago, shows the transition between dinosaurs and birds. The discovery was a major piece of evidence supporting Darwin's theory that one species evolves gradually into another. But Darwin's theory received a major setback in 1862 when British physicist **William Thomson** calculated the **age of Earth** from how fast it probably cooled since the time it was formed. The figure he came up with was no more than 400 million years, and possibly as short as 20 million years ago. Even 400 million years was not long enough for Darwin's gradual evolution to happen. Earth is now known to be closer to 4.5 billion years old.

In 1862, French chemist **Louis Pasteur** (1822–95) came closer to establishing that **microbes are responsible for many infectious diseases** with his studies of puerperal fever, an infection often caught by women



JAMES CLERK MAXWELL
(1831–79)

Born in Edinburgh, Scotland, British physicist James Clerk Maxwell laid the cornerstones of electromagnetic theory. With his brilliant math, he showed that electricity, magnetism, and light are all forms of electromagnetic fields. His four equations are the basis of all classical electromagnetic science.

during childbirth. Medicine was also advanced in 1864 when German physiologist **Felix Hoppe-Seyler** (1825–95) identified the **role of the iron-containing protein hemoglobin** in binding oxygen to red blood cells for transport through the blood. This was also the year that US president Abraham Lincoln signed the Yosemite Grant, which was the first step in creating California's now famous National Park in 1890.



Joseph Lister's introduction of antiseptic surgery with carbolic acid made operations far safer because the acid killed the germs that spread infection.

IN 1865, GERMAN PHYSICIST RUDOLF CLAUSIUS made an assertion about heat that was more significant than most people appreciated at the time. Starting from the second law of thermodynamics (heat flows only from hot places to cold), he came up with the **concept of entropy**. Entropy is a mathematical measure of the disorder in any system. To create order, heat needs to be concentrated, and this requires energy. So in any system, whether it is the human body or the entire Universe, entropy and disorder will increase **unless there is a continuous input of energy from outside** to maintain the concentration of heat. For example, life on Earth depends

“ THE ENERGY OF THE UNIVERSE IS CONSTANT. THE ENTROPY OF THE UNIVERSE TENDS TO A MAXIMUM. ”

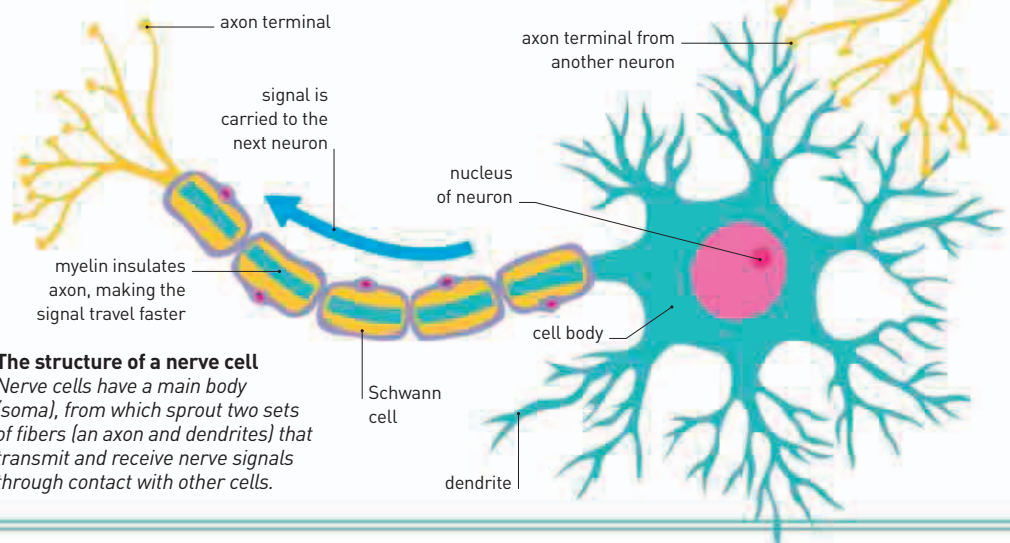
Rudolf Clausius, German physicist, from *On Various Forms of the Laws of Thermodynamics that are Convenient for Applications*, 1865

on the input of energy from the Sun; once the Sun's fuel has run out, this input will cease.

Another German, **Otto Friedrich Karl Deiters** (1834–63), observed the **basic features of nerve cells** for the first time under a microscope. He noted that each nerve cell has a main cell body, a long tail fiber (axon), and a

series of branching treelike fibers (dendrites).

In medicine, British surgeon **Joseph Lister** (1827–1912) pioneered the **use of carbolic acid as an antiseptic** during surgery, to clean instruments and wounds in order to reduce the chance of infection.



The structure of a nerve cell
Nerve cells have a main body (soma), from which sprout two sets of fibers (an axon and dendrites) that transmit and receive nerve signals through contact with other cells.

1864 Maxwell introduces the four key equations of electromagnetism

June 30, 1864 The Yosemite Grant establishes Yosemite National Park

August 20, 1864 British chemist John Newlands produces a periodic table of the elements

1864 William Thomson calculates the age of Earth

1864 Felix Hoppe-Seyler identifies the role of hemoglobin

Rudolf Clausius introduces the idea of entropy

Otto Deiters discovers the basic features of nerve cells

Joseph Lister pioneers antiseptic surgery

German chemist Friedrich Kekulé proposes a ring structure for a benzene molecule